INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
SALT LAKE CAMPUS

Degree: **B.C.A**, Stream: **B.C.A**

End Semester Examination 2023 - First Semester Examination, December-2023

Paper Code: BCAMD101

Paper Name: Digital Electronics

Full Marks: 100 Duration: 3 Hours

Part - A

Attempt 10 questions

Each question carries 2 Marks (2 X 10)

1. Convert the octal number 623.77 to hexadecimal. 2

or

1. Convert $(6734)_{10}$ to BCD form. 2

2. Convert $(199.3)_{10}$ to binary. 2

or

2. Find the 2's complement of $(10000)_2$. 2

3. What are Karnaugh maps, and how are they used in combinational logic simplification? 2

or

3. Explain the term "Universal Gates". 2

4. Calculate and simplify the expression (A+B)(AB)'. 2

or

4. What is a combinational logic circuit? 2

5. What is a shift register? 2

or

5. Infer the number of phases a Mod 6 counter passes through. 2

6. What is the basic difference between a Johnson counter and a ring counter? 2

or

6. Analyse how many states are there to count in Mod-10 counter? 2

7. Subtract $(110111)_2$ - $(11011)_2$ using 1's complement. 2

or

7. Convert the decimal number 250.5 to base 3. 2

8. What is de-multiplexer? 2

or

8. Find the 9's complement of $(13579)_{10}$. 2

9. The Boolean expression Y=AB' is logically equivalent to which logic gate? 2

or

9. What is a half adder, and how does it differ from a full adder? 2

10. Draw an OR logic using only NAND gates. 2

or

10. Can you explain the concept of a half adder and a full adder in combinational logic? 2

Part - B

Attempt 6 questions

Each question carries 5 Marks (5 X 6)

11. Multiply and evaluate the following numbers without converting it to decimal: (1230) base 4 and (23) base 4. 5

or

11. Perform the following 5

a) $(1011.0111)_2 + (1001.111)_2$

b) $(1101)_2 \times (1101)_2$

c) $(110110)_2 - (11001)_2$

d) $(11101)_2 / (1100)_2$

12. Evaluate a full-adder using only basic gates. 5

or

12. Infer the following expression to sum-of-product form (A + B) (B' + C) (A' + C). 5

13. Design a JK flipflop from T flipflop. 5

or

13. Implement the expression with a multiplexer F = Σ (0,1,3,4,8,9,15). 5

14. Write a note on Johnson counter. 5

or

14. Classify why the ordering of LSB and MSB are important for representing counter output? 5

15. Evaluate the 7's complement of 76 base 8. 5

or

15. Add and evaluate the following numbers without converting to decimal: (135.4)base 6 and (43.2)base 6. 5

16. Realise the logic expression Y = (A+B)(A'+C)(B+D) using basic gates. 5

or

16. What is meant by Decoder? Explain 3-to-8 line decoder with diagram and truth table. 5

Part - C

Attempt 5 questions

Each question carries 10 Marks (10 X 5)

17. Perform BCD subtraction 10

a) $(367)_{10} - (715)_{10}$

b) $(1230)_{10} - (23)_{10}$

or

17. Perform the following 10

A. $(317)_{10} + (489)_{10}$ BCD addition
B. 11001 – 111 (using 2s complement)
C. 111001 – 1101 (using 1s complement)
D. $(5247)_8$ to binary and Hexadecimal

18. Illustrate and simplify the Boolean expression: ABCD + A'B'CD + A'B. 10

or

18. Perform the following using 2s Complement method (do not use BCD subtraction) 10

a. $(+49)_{10} + (-37)_{10}$

b. $(-32)_{10} - (-16)_{10}$

19. Describe the working of Master-Slave JK flip-flop with circuit diagram and truth table. 10

or

19. Create the logic of the Parity Generator and Checker with the help of the logical circuit diagram. 10

20. Design a sequence generator for generating a sequence of 10110. 10

or

20. Explain the working of parallel-in-serial-out shift register with logic diagram. 10

21. Prove that the Gray code of 457 is 100101101. 10

or

21. Design the even parity code and odd parity bit code for 3-bit binary numbers. 10